Sayan Debnath

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SUMMARY

Software Engineer with nearly 2 years of experience building full-stack web applications and AI-powered systems. Experienced in Python, FastAPI, React, REST APIs, RAG pipelines, vector search, LLM integration, document processing, and cloud-based application development. Strong foundation in software engineering, backend development, databases, and AI application architecture.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL

Frontend: React.js, HTML, CSS, Tailwind CSS

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets

AI / Machine Learning: LLM Applications, RAG, LangChain, LangGraph, NLP, Prompt Engineering, Semantic Search, Vector Search, FAISS, Ollama

Databases: MongoDB, PostgreSQL, Redis

Cloud / DevOps: AWS (S3, EC2, Lambda), GCP, Docker, Git, CI/CD

Tools: Postman, Playwright

EXPERIENCE

Smarter Codes *Bangalore, Karnataka, India*
Full Stack Engineer Aug 2025 – May 2026

- Designed and developed an AI research engine using Python, FastAPI, vector similarity search reducing document retrieval time by 40%.
- Developed 3+ React UI modules for AI document pipeline monitoring and real-time tracking of 20+ document processing events per day.
- Developed backend APIs and integrated AI services to support document ingestion, retrieval, processing, and user-facing workflows.
- Worked with Docker, Redis, databases, and cloud services to support reliable development and deployment of production applications.

Cogzin Technologies *Siliguri, West Bengal, India*
Full Stack Developer July 2024 – Aug 2025

- Developed and maintained full-stack web applications using React.js, JavaScript, backend APIs, and database-driven workflows.
- Designed and integrated core platform modules including investor profiling, startup listings, recommendation workflows, and analytics dashboards.
- Improved data collection reliability and system integration by optimizing backend processes, reducing data sync errors by 30%
- Built reusable React.js components and integrated them with REST APIs to deliver responsive investor and startup-facing interfaces.

PROJECTS

DealHunter AI [GitHub Link] *Sep 2026 – Oct 2026*

- Agentic AI shopping assistant for product search, comparison, and personalized recommendations using live data.
- Integrated Model Context Protocol (MCP) tools for web product discovery, product-page inspection, and automated product comparison.
- Developed FastAPI backend services and a Next.js interface with Ollama-based local LLM inference for interactive product discovery.
- **Technologies:** Python, FastAPI, Next.js, TypeScript, Ollama, MCP, Agentic AI, Web Scraping

StreamFlix Recommendation System [GitHub Link] *Aug 2026 – Sep 2026*

- Built an AI-powered movie discovery platform with semantic search using natural-language queries.
- Engineered a scalable backend using Go, Gin, GORM, PostgreSQL, and pgvector, integrating a Python FastAPI embedding service with Sentence Transformers for 384-dimensional vector embeddings.
- Imported and processed 45,000+ movie records from TMDB and implemented vector similarity search for retrieval.
- **Technologies:** Go, Gin, GORM, PostgreSQL, pgvector, Python, FastAPI, Sentence Transformers, Next.js, Docker

EDUCATION

Indira Gandhi National Open University (IGNOU) *Maidan Garhi, Delhi, India*
Bachelor of Computer Applications (BCA) 2022-2025